CS 540 Spring 2024 Final

Exam Version: 1

First, Last Name:

Section:

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1. You cannot sit next (front, back, left (within two seats including an empty seat), right (within two seats including an empty seat)) to someone you know. Switch seats now if that's the case.

2. Fill in these fields (left to right) on the scantron sheet using a pencil:

   - LAST NAME (family name) and FIRST NAME (given name), fill in bubbles
   - IDENTIFICATION NUMBER is your Campus ID number, fill in bubbles
   - Under A, B, C of SPECIAL CODES, fill in your three-digit section number (001, 002, 003)
   - **Under F of SPECIAL CODES, fill in Exam version above.** This is very important!

   Make sure you fill all the above accurately in order to get graded.

3. Mark your answers on the Scantron sheet **and this handout.** Use a #2 pencil to mark all answers. Each question has exactly one correct answer.

4. You may only reference your one-sheet notes. Please turn off and put away portable electronics now.

5. If you need to use the restroom, your phone must remain in the room, placed face down on your desk and kept visible to the proctors.

6. You have 120 minutes to take the exam. Once you have finished, please show your student ID to one of the proctors and submit both your scantron and this handout.

Good luck!

1. Calculate the convolution of matrix A with kernel matrix B, for no padding and a stride of 1.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 3 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

   A. $\begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$
   B. $\begin{bmatrix} 4 & 3 \\ 0 & 5 \end{bmatrix}$
   C. $\begin{bmatrix} 3 & 2 \\ 0 & 6 \end{bmatrix}$
   D. $\begin{bmatrix} 8 & 6 \\ 1 & 12 \end{bmatrix}$
   E. None of the above.

> **Solution:** B: Row 1, column 1: $1 * 1 + 1 * 3 = 4$. Row 1, column 2: $1 * 2 + 1 * 1 = 3$. Row 2, column 1: $1 * 0 + 1 * 0 = 0$. Row 2, column 2: $1 * 3 + 1 * 2 = 5$

2. What is the most common type of pooling within convolutional neural networks?
   A. Sinusoidal pooling
   B. Cumulative pooling
   C. Swim pooling
   D. Negative pooling
   E. None of the above.

> **Solution:** E: Max pooling and average pooling are the only pooling discussed in the slides.

3. Consider a convolution layer with 64 filters. Each filter has a size of 8 x 8 x 3, a stride of 3 x 3. Given an input image of size 11 x 11 x 3, if we don't allow a filter to fall outside of the input, what is the output size?
   A. 1 x 1 x 32
   B. 4 x 4 x 32
   C. 1 x 1 x 64
   D. 2 x 2 x 64

E. None of the above.

Solution: D. (11 - 8 + 3 + 0) / 3 = 2

4. Which of the following is a **not** reason why Convolutional Neural Networks(CNNs) are preferred over Multilayer Perceptrons (MLP - fully connected networks) for image processing?

   A. In convolutional layers, kernel weights are invariant to where the features appear in the image. This is not the case in dense layers.

   B. CNNs can learn non-linear features for complex images. But MLP cannot.

   C. The extracted features can be visualized by inspecting the activation maps.

   D. Convolutional layers assume that nearby pixels share similar structure.

   E. None of the above.

Solution: B. MLP can learn non-linear features for complex images as well.

5. Which of the below can you implement to solve the gradient vanishing problem?

   A. Reduce the batch size

   B. Increase the depth of network

   C. Oversample majority classes and downsample minority classes

   D. Use SGD optimization

   E. None of the above.

Solution: E

6. Which of the following is **not** a reason that adding more layers to a neural network decreases performance?

   A. Neural networks with more layers have more parameters which can lead overfitting

   B. Larger models have larger functions spaces and are harder to optimize

   C. Adding additional layers could cause vanishing gradients

   D. Additional layers could cause instability in the loss functions so they have unpredictable behavior

   E. None of the above.

Solution: D - adding additional layers to a neural network does not impact the behavior of the loss functions. A: additional layers means that a network will have more parameters to optimize over and could fit to noise. B: larger models have more parameters and can learn more possible functions, so it can be harder problem to optimize the learned result. C: with more layers, it is more likely that the backpropagated gradients become very small ie. vanish.

7. Why might a particular image transformation be unhelpful for data augmentation?
   A. A reflection can change the label of the image
   B. The contrast of the image might not be changed enough
   C. Cropping can remove too much of the original image
   D. The cropped image can be too similar to the original image
   E. All of the above

Solution: E - all of the above. A: could reflect a "d" to a "b". B and D: if not enough changes about the image, there is nothing new for the model to learn. C: if we want to identify the number of cats and in an image, cropping too much can remove the cats and dogs from an image and change the label.

8. Which if the following is **false** about regularization?
   A. Applying data augmentation blindly may result in useless or harmful training points.
   B. Data augmentation can be applied for text.
   C. Regularization always introduces complexity to a model to avoid overfitting.
   D. We can modify the loss function by adding a regularization term.
   E. The transformations for data augmentation depend on the domain.

Solution: C - Regularization is used to avoid overfitting, but in some interpretations enforces simplicity not complexity.

9. Which of the following about residual networks is **true**?
   A. Since learning zero can be hard, residual connections help learn the identity instead.
   B. The residual connections add paths to the computation path.
   C. Learning the identity means that adding more layers decreases performance.

D. ResNets have worse performance than CNNs that do not have skip connections.

E. None of the above.

> **Solution:** B - is true. A: learning the identity is hard, so residual connections learn zero instead. C: being able to learn the identity means that additional layers can't do worse. D. ResNets improve on traditional CNNs.

10. In Breadth First Search (BFS), which of the following data structures is typically used to keep track of the nodes to be visited next?

    A. Stack

    B. Queue

    C. Priority queue

    D. Heap

    E. None of the above.

> **Solution:** B - Breadth First Search (BFS) explores nodes in a graph level by level. To achieve this, it uses a queue data structure to keep track of the nodes to be visited next. In BFS, nodes are visited in the order they are discovered, and the FIFO (First-In-First-Out) property of a queue ensures that the nodes at the current level are explored before moving on to the nodes at the next level.

11. Which of the following graph traversal algorithms is most likely to continue indefinitely when applied to an infinite depth graph?

    A. Breadth First Search (BFS)

    B. Depth First Search (DFS)

    C. Uniform Cost Search

    D. Iterative Deepening Depth-First Search (IDDFS)

    E. None of the above.

> **Solution:** B - Depth First Search (DFS) explores as far as possible along each branch of the graph before backtracking. In an infinite depth graph, DFS can potentially continue indefinitely, as it will keep traversing deeper and deeper into the graph without bound. Since DFS does not have a built-in mechanism to detect or handle infinite depths, it may not terminate in such scenarios.

12. Consider a scenario where an uninformed search algorithm is used to find a route from a start city to a destination city on a map. Which of the following statements accurately describes the behavior of such an algorithm?

A. Uninformed search algorithms always choose the shortest path between cities, ensuring the most efficient route.

B. Uninformed search algorithms may traverse through longer routes before discovering the shortest path to the destination.

C. Uninformed search algorithms use additional domain-specific knowledge to prioritize exploration, ensuring optimality.

D. Uninformed search algorithms always find the optimal route regardless of the complexity of the map.

E. None of the above.

Solution: B - Uninformed search algorithms, such as Breadth First Search (BFS) and Depth First Search (DFS), explore the search space without using additional domain-specific knowledge or heuristics. In the context of finding routes on a map, uninformed search algorithms may traverse through longer routes or explore alternative paths before discovering the shortest path to the destination. This is because these algorithms systematically explore the search space without considering the lengths of individual edges or using heuristic information.

13. A leaky ReLU is an activation function with the definition $f(x) = \max(\alpha x, x)$. A standard ReLU is a special case of leaky ReLU with $\alpha = 0$. Now suppose $\alpha = 0.2$, and the input to the leaky ReLU is -0.5, what is the output of the leaky ReLU?

    A. -0.1

    B. -0.01

    C. 0.1

    D. 0

    E. None of the above.

Solution: A - By definition, we have that $f(x) = \max(0.2x, x)$, which implies that $f(-0.5) = \max(-0.1, -0.5) = -0.1$.

14. Consider a perceptron with weights $w_1 = 1, w_2 = 2, w_3 = 3$, bias $b = -10$, and ReLU activation. Given the input vector $x_1 = 1, x_2 = 2, x_3 = 3$, what will the perceptron output?

    A. 0

    B. 4

    C. 14

    D. 24

E. None of the above.

Solution: B - $\text{ReLU}(w_1x_1 + w_2x_2 + w_3x_3 + b) = \text{ReLU}(4) = \max(0, 4) = 4$.

15. Consider a simple neural network $y = \sigma(x^\intercal W + b^\intercal)$ with 3 inputs and two outputs and no hidden layers. Given input $x^\intercal = [3, 2, 1]$ with weight matrix $W = \begin{bmatrix} 0.1 & -0.2 \\ 0.2 & 0.3 \\ -0.3 & 0.4 \end{bmatrix}$, biases set to $b^\intercal = [0.1, 0.2]$, and a ReLU activation function on the outputs. What is the output of the network?

    A. $[0, 0]$

    B. $[0.5, 0]$

    C. $[0, 0.6]$

    D. $[0.5, 0.6]$

    E. None of the above.

Solution: D - By definition of matrix multiplication, we have that $\text{ReLU}(x^\intercal W + b^\intercal) = \text{ReLU}([0.5, 0.6]) = [0.5, 0.6]$.

16. Consider a search space S that consists of the integers: $\ldots, -3, -2, -1, 0, 1, 2, \ldots$. There is a goal state g (an integer). The successor states for any state s are $\{s - 1, s + 1\}$, and the cost to go to either successor is 1.

    Suppose we use A^* search to solve our search problem. Select the following heuristic functions that can be used:

    A. $h(s) = g - s$,

    B. $h(s) = \frac{1}{2}|s - g|$,

    C. $h(s) = \max(s - g, \frac{1}{4})$,

    D. $h(s) = \min(-1, |s - g|, e^{s-g}, (s - g)^2)$,

    E. None of the above.

Solution: B. Others have cases that are negative.

17. Two firms, A and B, are deciding whether to launch a new product. Each firm can either launch or not launch. Their profits depend on their choices, and the payoff matrix is as follows:

	B: Launch	B: Not Launch
A: Launch	$(20, 20)$	$(40, 10)$
A: Not Launch	$(10, 40)$	$(30, 30)$

What is the strictly dominant strategy for each firm?

A. A's dominant strategy is to launch, and B's dominant strategy is not to launch.

B. A's dominant strategy is to launch, and B's dominant strategy is to launch.

C. A's dominant strategy is not to launch, and B's dominant strategy is to launch.

D. A's dominant strategy is not to launch, and B's dominant strategy is not to launch.

E. None of the above.

Solution: B. - If B launches, then A is getting the maximum payoff from launching, which is 20. If B does not launch then A is getting maximum payoff again if they launch which is 40. If A launches, then for B is also better to launch because the payoff will be 20. If A does not launch, B is getting maximum payoff if B launches, which is 40. Thus, for both player the dominant strategy is to launch.

18. Which of the following statements is true about Nash equilibrium?

A. The mixed strategy equilibrium for the game Rock-Paper-Scissors is when both players play the mixed strategy that puts equal probabilities on all three actions.

B. In a pure strategy Nash equilibrium, players can deviate from their strategies to increase their payoff, while in a mixed strategy Nash equilibrium, any deviation would result in a lower payoff.

C. A mixed strategy Nash equilibrium occurs when all players choose a specific action with certainty, while a pure strategy Nash equilibrium occurs when all players choose their actions deterministically.

D. Both pure and mixed strategy Nash equilibria involve certain level of uncertainty in the choice of strategies.

E. None of the above.

Solution: A. Pure Nash equilibrium is deterministic while mixed strategy Nash equilibrium has certain randomness.

19. Which of the following statements is true about the minimax algorithm?

A. The minimax algorithm can be used to find the optimal strategy for a single-player game as it aims to maximize the player's score.

B. The minimax algorithm finds the optimal strategy for each player in a two-player cooperative game, aiming to maximize the combined score of both players.

C. The time complexity of a minimax algorithm is $O(bm)$.

D. The minimax algorithm is a technique used to minimize the maximum possible loss in a two-player zero-sum game, where each player tries to maximize their score while minimizing their opponent's score.

E. None of the above.

Solution: D. A and B describe the objective of the minimax algorithm wrong. The time complexity of a minimax algorithm is $O(b^m)$.

20. In a two-player game, each player has three possible moves in a single round. One round is considered finished after both players have played. The game lasted for 4 rounds in total. Considering this game scenario, how many leaf nodes are there in the corresponding minimax tree?

A. 3^4

B. 3^8

C. $\sum_{k=0}^{8} 3^k$

D. $3^2 \times 4$

E. None of the above.

Solution: B. The tree has a branching factor of 3 and a depth of 8.

21. Which of the following domain is *best* suited for Reinforcement Learning as opposed to other kinds of machine learning?

A. Learning a translation between two languages.

B. Transcribing (speech to text) audio from a recording

C. Generating images of dogs from text input.

D. Controlling a helicopter while it flies around a city.

E. None of the above.

Solution: D - This is a sequential decision making problem which RL is optimal for.

22. We are choosing between two discount factors for our reinforcement learning problem. We want the agent to care significantly more about the *present* than the *future*. Should we choose $\gamma = 0.25$ or $\gamma = 0.75$ and why?

   A. $\gamma = 0.25$ Because lower discount factors increase importance of the present

   B. $\gamma = 0.25$ Because lower discount factors increase importance of the future

   C. $\gamma = 0.75$ Because higher discount factors increase importance of the present

   D. $\gamma = 0.75$ Because higher discount factors increase importance of the future

   E. None of the above.

> **Solution:** A - as n increases where discounted future reward $= \gamma = 0.25^n$, future reward will go to zero much faster as opposed to $\gamma = 0.75^n$ so the present is higher weighted for $\gamma = 0.25$.

23. In a Markov Decision Process, what does the Markov property refer to?

   A. The probability of transitioning to a particular state depends only on the current state and action

   B. The probability of transitioning to a particular state depends on the entire history of states and actions

   C. The reward for an action depends on the next state and all previous states

   D. The state transitions occur deterministically

   E. None of the above.

> **Solution:** A
>
> Markov property means the current state (including the transition probabilities) only depend on the current state and action.

24. What is NOT true about Q-learning with ϵ-greedy exploration where $\epsilon = 0.1$?

   A. At every step, the agent selects the action with the highest Q-value with probability $1 - \epsilon$.

   B. The point of a non-zero ϵ is to encourage exploration.

   C. The agent selects a random action with probability ϵ.

   D. The policy can achieve the maximum possible reward.

   E. None of the above.

Solution: D - Since at every time step there is a chance of taking a suboptimal action due to epsilon-greedy, the policy cannot achieve the maximum reward on a task.

25. Supposed you have the following information about an environment:

    1. The discount factor is 0.8
    2. The reward in s_1 taking action a_1 is 3
    3. The transition probabilities are: $P(s_2|s_1, a_1) = 0.6$ and $P(s_3|s_1, a_1) = 0.4$
    4. Currently, $V(s_2) = 10$ and $V(s_3) = 6$

    Remember the update for value iteration is $V_{t+1}(s) = r(s) + \gamma \max_a \sum_{s'} P(s'|s, a)V(s')$. After a single iteration of value iteration, what is the value for state s_1 (what is $V(s_1)$)? Choose the **closest** option.

    A. 8

    B. 10

    C. 12

    D. 14

    E. None of the above.

Solution: B

$V(s_1) = 3 + 0.8(0.6 \times 10 + 0.4 \times 6) = 9.72 \ = 10$